MEPCO SCHLENK ENGINEERING COLLEGE (AUTONOMOUS), SIVAKASI

Department of Computer Science and Engineering

19CSP01 – Free Open Source Software

Assignment 3: Web Application Development using MySQL (Backend) and PHP (Frontend)

Question

Develop a simple Web application of your choice using:

- MySQL as back-end
- PHP or Python as front-end

Aim

To design and develop a full-stack Campus Task Planner (CTP) web application utilizing MySQL as the relational database backend and PHP as the server-side frontend, incorporating 4 relational tables (categories, priorities, tasks, task_checklists) and implementing complete Create, Read, Update, and Delete (CRUD) operations with server-side input validation and dynamic user feedback.

Introduction

Web applications built upon Free and Open Source Software (FOSS) technologies form the foundation of modern internet infrastructure. The LAMP stack (Linux, Apache, MySQL, PHP) represents one of the most widely adopted open-source architectural patterns for dynamic web development.

PHP (Hypertext Preprocessor) is an open-source, server-side scripting language engineered specifically for web development, offering seamless integration with relational databases, dynamic session management, and robust request handling. MySQL is an enterprise-grade, open-source Relational Database Management System (RDBMS) that provides ACID compliance, foreign key relational integrity, and high-performance querying.

In this assignment, a complete Campus Task Planner is constructed. Rather than using an isolated single table, the backend is engineered with 4 interconnected tables structured via Primary Key and Foreign Key constraints, reflecting real-world institutional database architecture.

System Requirements

- Operating System: Windows 11 (64-bit)
- Web Server: Apache HTTP Server 2.4.58 (via XAMPP 8.2.12)
- Database Management System: MySQL / MariaDB 10.4.32 (via phpMyAdmin 5.2.1)
- Server-Side Language: PHP Version 8.2.12
- Frontend Technologies: HTML5, CSS3, JavaScript (Native, Zero External Dependencies)
- Web Browser: Google Chrome / Microsoft Edge

Database Architecture and Relational Schema

The database 'foss_task_planner' comprises 4 relational tables establishing 1-to-Many relationships:

- categories: Entity table storing category codes, names, and academic descriptions (Primary Key: category_id).
- priorities: Entity table maintaining priority levels, urgency labels, and color codes (Primary Key: priority_id).
- tasks: Entity table recording task details and linking to categories and priorities via Foreign Key constraints (Primary Key: task_id, Foreign Keys: category_id, priority_id).
- task_checklists: Relational associative table mapping tasks to checklist items along with boolean completion states (Primary Key: checklist_id, Foreign Key: task_id).

The detailed schema definition for each table is tabulated below:

Table Name	Field Name	Data Type & Constraints	Role / Description
categories	category_id	INT AUTO_INCREMENT PRIMARY KEY	Unique Category ID
categories	category_code	VARCHAR(15) NOT NULL UNIQUE	Category Abbreviation (e.g. ACAD)
categories	category_name	VARCHAR(60) NOT NULL	Full Category Title
categories	description	VARCHAR(150)	Academic Scope Description
priorities	priority_id	INT AUTO_INCREMENT PRIMARY KEY	Unique Priority ID
priorities	priority_level	INT NOT NULL UNIQUE	Numeric Urgency (1=High, 2=Med, 3=Low)
priorities	priority_label	VARCHAR(20) NOT NULL	Priority Name (High, Medium, Low)
priorities	badge_color	VARCHAR(20) NOT NULL	UI Badge Hex Color Code
tasks	task_id	INT AUTO_INCREMENT PRIMARY KEY	Unique Task ID
tasks	task_title	VARCHAR(150) NOT NULL	Task Title / Subject

tasks	description	TEXT	Detailed Task Specifications
tasks	due_date	DATE NULL	Task Due Date
tasks	task_status	ENUM('Pending','In Progress',...)	Current Execution Status
tasks	category_id	INT NOT NULL, FOREIGN KEY	FK referencing categories(category_id)
tasks	priority_id	INT NOT NULL, FOREIGN KEY	FK referencing priorities(priority_id)
tasks	created_at	TIMESTAMP DEFAULT NOW	Creation Timestamp
task_checklists	checklist_id	INT AUTO_INCREMENT PRIMARY KEY	Unique Checklist Item ID
task_checklists	task_id	INT NOT NULL, FOREIGN KEY	FK referencing tasks(task_id)
task_checklists	item_title	VARCHAR(150) NOT NULL	Checklist Item Description
task_checklists	is_completed	TINYINT(1) DEFAULT 0	Completion State (0=No, 1=Yes)

Entity-Relationship (ER) Diagram: Chen's Model

The logical structure of the 4-table relational database is formalized using Chen's notation. Rectangles designate Entities (CATEGORIES, PRIORITIES, TASKS, TASK_CHECKLISTS), diamonds represent Relationships (CLASSIFIES, ASSIGNED_TO, CONTAINS), and ovals indicate attributes with Primary Keys underlined:

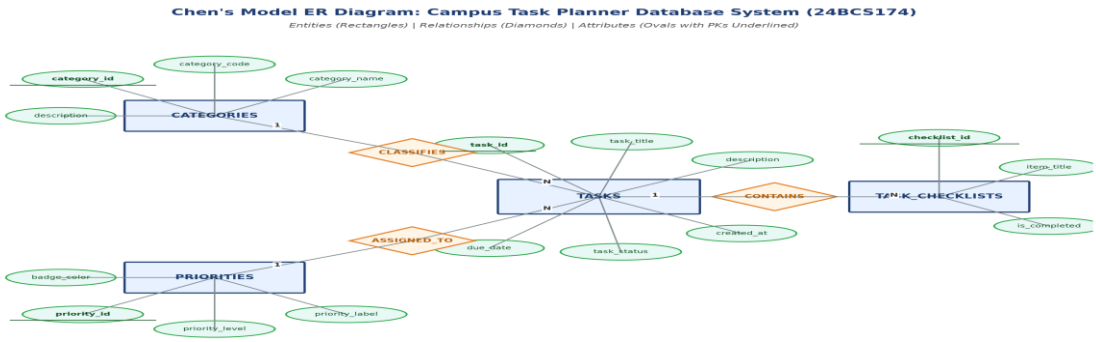


Figure 1: Chen's Model ER Diagram (Entities, Relationships, Attributes, and Cardinalities)

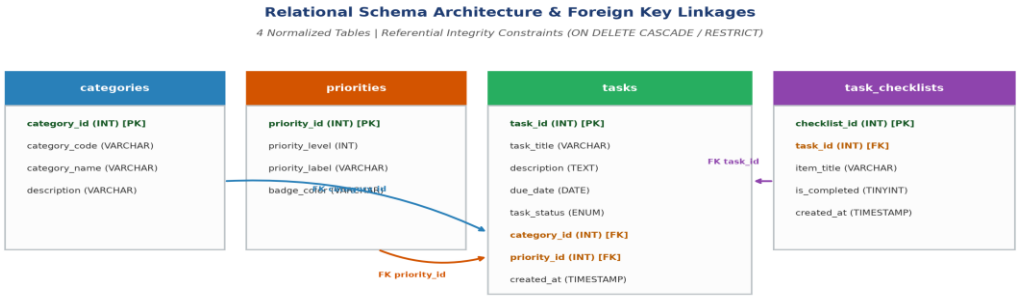


Figure 1b: Relational Schema Architecture Overview

Complete Source Code Implementation

Below are the complete, unmodified source codes for the backend database definitions, database connectivity, and full frontend user interface scripts with embedded PHP server-side logic:

1. Database Schema and Seed Data Script (schema.sql)

```
-- schema.sql: 4 Relational Tables for Campus Task Planner System
-- Developed for 19CSP01 - Free Open Source Software
-- Student: Ragul Ramachandran (24BCS174)

CREATE DATABASE IF NOT EXISTS foss_task_planner;
USE foss_task_planner;

-- Table 1: Categories Entity
CREATE TABLE IF NOT EXISTS categories (
    category_id     INT AUTO_INCREMENT PRIMARY KEY,
    category_code   VARCHAR(15)  NOT NULL UNIQUE,
    category_name   VARCHAR(60)  NOT NULL,
    description     VARCHAR(150)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Table 2: Priorities Entity
CREATE TABLE IF NOT EXISTS priorities (
    priority_id     INT AUTO_INCREMENT PRIMARY KEY,
    priority_level  INT         NOT NULL UNIQUE,
    priority_label  VARCHAR(20)  NOT NULL,
    badge_color     VARCHAR(20)  NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Table 3: Tasks Entity (Foreign Keys -> categories, priorities)
CREATE TABLE IF NOT EXISTS tasks (
    task_id         INT AUTO_INCREMENT PRIMARY KEY,
    task_title      VARCHAR(150) NOT NULL,
    description     TEXT,
    due_date        DATE        NULL,
    task_status     ENUM('Pending', 'In Progress', 'Completed') NOT NULL DEFAULT 'Pending',
    category_id     INT         NOT NULL,
    priority_id     INT         NOT NULL,
    created_at      TIMESTAMP    DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (category_id) REFERENCES categories(category_id) ON DELETE CASCADE,
    FOREIGN KEY (priority_id) REFERENCES priorities(priority_id) ON DELETE RESTRICT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

```
-- Table 4: Task Checklists / Subtasks (Foreign Key -> tasks)
CREATE TABLE IF NOT EXISTS task_checklists (
    checklist_id    INT AUTO_INCREMENT PRIMARY KEY,
    task_id         INT          NOT NULL,
    item_title      VARCHAR(150) NOT NULL,
    is_completed    TINYINT(1)   DEFAULT 0,
    created_at      TIMESTAMP    DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (task_id) REFERENCES tasks(task_id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Seed Data Insertion
INSERT INTO categories (category_code, category_name, description) VALUES
('ACAD', 'Academic & Coursework', 'Assignments, lecture notes, and lab preparations'),
('PROJ', 'Mini Project & Capstone', 'Software engineering design, coding, and deliverables'),
('PERS', 'Personal & Skill Development', 'Open-source contributions, webinars, and reading'),
('EXAM', 'Examination Preparation', 'Internal assessments, semester review, and question banks');

INSERT INTO priorities (priority_level, priority_label, badge_color) VALUES
(1, 'High', '#dc3545'),
(2, 'Medium', '#fd7e14'),
(3, 'Low', '#20c997');

INSERT INTO tasks (task_title, description, due_date, task_status, category_id, priority_id) VALUES
('Complete FOSS Assignment 2 & 3', 'Develop Python package and full-stack PHP/MySQL CRUD application with report.', '2026-10-23', 'In Progress', 1, 1),
('Review Operating Systems IPC Lab', 'Study message queues and shared memory implementation for upcoming lab exam.', '2026-10-25', 'Pending', 1, 2),
('FOSS Open Source Contribution', 'Fork active repository, create descriptive feature pull request, and document diff.', '2026-10-28', 'Pending', 3, 2),
('Organise Study Materials for SEM V', 'Consolidate lecture presentations and reference textbooks into cloud drive.', '2026-10-21', 'Completed', 1, 3);

INSERT INTO task_checklists (task_id, item_title, is_completed) VALUES
(1, 'Author Python package with setup.py and test suite', 1),
(1, 'Implement 4-table relational MySQL schema and PHP backend', 1),
(1, 'Generate documentation and commit to GitHub', 0),
(2, 'Test two-way pipe implementation', 1),
(2, 'Verify message queue synchronization', 0),
(3, 'Clone target repository locally', 1),
(3, 'Submit pull request with passing tests', 0),
(4, 'Backup database dumps and configuration files', 1);
```

2. Database Connection Script (db.php)

```
<?php
/**
 * db.php - Database Configuration and Connection Handler
 * Student: Ragul Ramachandran (24BCS174)
 * Course: 19CSP01 - Free Open Source Software
```

```
 */

$host     = "127.0.0.1";
$user     = "root";
$password = "";
$database = "foss_task_planner";
$port     = (int) (getenv("TASK_DB_PORT") ?: 3306);

// Establish database connection using mysqli
$conn = @new mysqli($host, $user, $password, $database, $port);

if ($conn->connect_error) {
    die("<div style='font-family:sans-serif;padding:20px;color:#721c24;background:#f8d7da;border:1px solid
#f5c6cb;border-radius:6px;max-width:600px;margin:40px auto;'>"
        . "<strong>Database Connection Error:</strong> " . htmlspecialchars($conn->connect_error)
        . "<br><small>Please verify that MySQL service is running and credentials are
accurate.</small></div>");
}

$conn->set_charset("utf8mb4");
?>
```

3. Home Dashboard and Read Operation Script (index.php)

```
<?php
/**
 * index.php - Main Dashboard (Read Operation & Statistics)
 * Student: Ragul Ramachandran (24BCS174)
 * Course: 19CSP01 - Free Open Source Software
 */
require_once "db.php";

// Handle quick status toggle
if (isset($_GET["toggle"]) && (int)$_GET["toggle"] > 0) {
    $tid = (int)$_GET["toggle"];
    $stmt = $conn->prepare("UPDATE tasks SET task_status = IF(task_status = 'Completed', 'Pending',
'Completed') WHERE task_id = ?");
    $stmt->bind_param("i", $tid);
    $stmt->execute();
    header("Location: index.php?msg=toggled");
    exit;
}

// Fetch filter parameters
$filter_category = isset($_GET["cat"]) ? (int)$_GET["cat"] : 0;
$filter_status   = isset($_GET["status"]) ? trim($_GET["status"]) : "";
$search_query    = isset($_GET["q"]) ? trim($_GET["q"]) : "";
```

```
// Build relational query joining 4 tables
$sql = "SELECT
          t.task_id,
          t.task_title,
          t.description,
          t.due_date,
          t.task_status,
          t.created_at,
          c.category_name,
          c.category_code,
          p.priority_label,
          p.badge_color,
          GROUP_CONCAT(tc.item_title SEPARATOR ' • ') AS checklist_summary,
          COUNT(tc.checklist_id) AS total_subtasks,
          SUM(IF(tc.is_completed = 1, 1, 0)) AS completed_subtasks
       FROM tasks t
       JOIN categories c ON t.category_id = c.category_id
       JOIN priorities p ON t.priority_id = p.priority_id
       LEFT JOIN task_checklists tc ON t.task_id = tc.task_id
       WHERE 1=1 ";

$params = [];
$types  = "";

if ($filter_category > 0) {
   $sql .= "AND t.category_id = ? ";
   $params[] = $filter_category;
   $types   .= "i";
}
if (!empty($filter_status)) {
   $sql .= "AND t.task_status = ? ";
   $params[] = $filter_status;
   $types   .= "s";
}
if (!empty($search_query)) {
   $sql .= "AND (t.task_title LIKE ? OR t.description LIKE ?) ";
   $like = "%" . $search_query . "%";
   $params[] = $like;
   $params[] = $like;
   $types   .= "ss";
}

$sql .= "GROUP BY t.task_id ORDER BY t.due_date ASC, p.priority_level ASC";
```

```
$stmt = $conn->prepare($sql);
if (!empty($params)) {
    $stmt->bind_param($types, ...$params);
}
$stmt->execute();
$tasks_result = $stmt->get_result();

// Overall KPI Statistics
$kpi_total     = $conn->query("SELECT COUNT(*) AS c FROM tasks")->fetch_assoc()["c"];
$kpi_pending   = $conn->query("SELECT COUNT(*) AS c FROM tasks WHERE task_status = 'Pending'")->fetch_assoc()["c"];
$kpi_progress  = $conn->query("SELECT COUNT(*) AS c FROM tasks WHERE task_status = 'In Progress'")->fetch_assoc()["c"];
$kpi_completed = $conn->query("SELECT COUNT(*) AS c FROM tasks WHERE task_status = 'Completed'")->fetch_assoc()["c"];
$kpi_high      = $conn->query("SELECT COUNT(*) AS c FROM tasks t JOIN priorities p ON t.priority_id = p.priority_id WHERE p.priority_label = 'High'")->fetch_assoc()["c"];

// Categories for filter dropdown
$categories_res = $conn->query("SELECT * FROM categories ORDER BY category_name ASC");

// Feedback message handler
$feedback = "";
if (isset($_GET["msg"])) {
    $m = $_GET["msg"];
    if ($m === "created") {
        $feedback = "<div class='alert alert-success'><strong>Success!</strong> Task record created successfully with associated category and priority.</div>";
    } elseif ($m === "updated") {
        $feedback = "<div class='alert alert-info'><strong>Updated!</strong> Task details and checklist items updated successfully.</div>";
    } elseif ($m === "deleted") {
        $feedback = "<div class='alert alert-danger'><strong>Deleted!</strong> Task record and associated checklist removed from database.</div>";
    } elseif ($m === "toggled") {
        $feedback = "<div class='alert alert-success'><strong>Status Changed!</strong> Task status was toggled successfully.</div>";
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Campus Task Planner - FOSS WebApp (24BCS174)</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header class="app-header">
    <div class="header-container">
```

```
      <div class="brand">
        <h1> 📋 Campus Task Planner</h1>
        <p class="subtitle">19CSP01 – Free Open Source Software | Relational CRUD WebApp | Register No:
24BCS174</p>
      </div>
      <div class="header-badge">
        <span class="badge badge-tech">MySQL 8.x + PHP 8.2</span>
        <span class="badge badge-rel">4 Relational Tables</span>
      </div>
    </div>
</header>

<main class="container">
    <?= $feedback ?>

    <!-- KPI Metric Cards -->
    <section class="kpi-grid">
      <div class="kpi-card card-blue">
        <div class="kpi-val"><?= $kpi_total ?></div>
        <div class="kpi-lbl">Total Tasks</div>
      </div>
      <div class="kpi-card card-orange">
        <div class="kpi-val"><?= $kpi_pending ?></div>
        <div class="kpi-lbl">Pending</div>
      </div>
      <div class="kpi-card card-cyan">
        <div class="kpi-val"><?= $kpi_progress ?></div>
        <div class="kpi-lbl">In Progress</div>
      </div>
      <div class="kpi-card card-green">
        <div class="kpi-val"><?= $kpi_completed ?></div>
        <div class="kpi-lbl">Completed</div>
      </div>
      <div class="kpi-card card-red">
        <div class="kpi-val"><?= $kpi_high ?></div>
        <div class="kpi-lbl">High Priority</div>
      </div>
    </section>

    <!-- Toolbar: Search, Filter, and Add New Task -->
    <div class="toolbar-card">
      <form method="GET" action="index.php" class="filter-form">
        <input type="text" name="q" placeholder="Search tasks..." value="<?=
htmlspecialchars($search_query) ?>" class="input-search">

        <select name="cat" class="input-select">
```

```
            <option value="0">All Categories</option>
            <?php while ($cat = $categories_res->fetch_assoc()): ?>
                <option value="<?= $cat['category_id'] ?>" <?= $filter_category == $cat['category_id'] ?
'selected' : '' ?>>
                    <?= htmlspecialchars($cat['category_name']) ?>
                </option>
            <?php endwhile; ?>
        </select>

        <select name="status" class="input-select">
            <option value="">All Statuses</option>
            <option value="Pending" <?= $filter_status === 'Pending' ? 'selected' : '' ?>>Pending</option>
            <option value="In Progress" <?= $filter_status === 'In Progress' ? 'selected' : '' ?>>In
Progress</option>
            <option value="Completed" <?= $filter_status === 'Completed' ? 'selected' : ''
?>>Completed</option>
        </select>

        <button type="submit" class="btn btn-filter">Filter</button>
        <?php if ($filter_category > 0 || !empty($filter_status) || !empty($search_query)): ?>
            <a href="index.php" class="btn btn-clear">Reset</a>
        <?php endif; ?>
    </form>

    <a href="task_form.php" class="btn btn-add">➕ Add New Task</a>
</div>

<!-- Tasks Data Table -->
<div class="table-card">
    <table class="data-table">
        <thead>
            <tr>
                <th style="width: 5%;">#</th>
                <th style="width: 25%;">Task Title & Description</th>
                <th style="width: 15%;">Category</th>
                <th style="width: 10%;">Priority</th>
                <th style="width: 12%;">Due Date</th>
                <th style="width: 11%;">Status</th>
                <th style="width: 12%;">Subtasks</th>
                <th style="width: 10%; text-align: center;">Actions</th>
            </tr>
        </thead>
        <tbody>
            <?php if ($tasks_result->num_rows === 0): ?>
                <tr>
                    <td colspan="8" class="text-center empty-state">No matching tasks found. Click "Add New
Task" to create one.</td>
                </tr>
```

```
        <?php else: ?>
            <?php while ($row = $tasks_result->fetch_assoc()): ?>
                <tr>
                    <td><span class="task-id-badge"><?= $row["task_id"] ?></span></td>
                    <td>
                        <strong><?= htmlspecialchars($row["task_title"]) ?></strong>
                        <?php if (!empty($row["description"])): ?>
                            <div class="task-desc"><?= htmlspecialchars($row["description"]) ?></div>
                        <?php endif; ?>
                    </td>
                    <td>
                        <span class="badge badge-cat"><?= htmlspecialchars($row["category_name"])
?></span>
                    </td>
                    <td>
                        <span class="badge" style="background-color: <?= $row['badge_color'] ?>; color:
#fff;">
                            <?= htmlspecialchars($row["priority_label"]) ?>
                        </span>
                    </td>
                    <td><?= $row["due_date"] ? htmlspecialchars($row["due_date"]) : "<span class='text-
muted'>None</span>" ?></td>
                    <td>
                        <a href="index.php?toggle=<?= $row['task_id'] ?>" title="Click to toggle status"
class="status-pill status-<?= strtolower(str_replace(' ', '-', $row['task_status'])) ?>">
                            <?= htmlspecialchars($row["task_status"]) ?>
                        </a>
                    </td>
                    <td>
                        <?php if ($row["total_subtasks"] > 0): ?>
                            <div class="subtask-progress">
                                <span class="subtask-ratio"><?= $row["completed_subtasks"] ?>/<?=
$row["total_subtasks"] ?> Done</span>
                            </div>
                            <small class="checklist-text" title="<?= htmlspecialchars($row['checklist_summary'])
?>">
                                <?= htmlspecialchars(mb_strimwidth($row["checklist_summary"], 0, 32, "...")) ?>
                            </small>
                        <?php else: ?>
                            <span class="text-muted">-</span>
                        <?php endif; ?>
                    </td>
                    <td class="action-cell">
                        <a href="task_form.php?id=<?= $row['task_id'] ?>" class="btn-action btn-edit"
title="Edit Task">Edit</a>
                        <a href="delete_task.php?id=<?= $row['task_id'] ?>" class="btn-action btn-delete"
onclick="return confirm('Are you sure you want to permanently delete this task?');" title="Delete
Task">Delete</a>
                    </td>
                </tr>
            <?php endwhile; ?>
        <?php endif; ?>
    </tbody>
```

```
        </table>
    </div>
</main>

<footer class="app-footer">
    <p>Campus Task Planner Web Application | Built with PHP, MySQL & CSS3 | 19CSP01 Assignment 3 |
Ragul Ramachandran (24BCS174)</p>
</footer>

</body>
</html>
```

4. Task Creation and Updation Form Script (task_form.php)

```
<?php
/**
 * task_form.php - Create & Update Operation Form
 * Student: Ragul Ramachandran (24BCS174)
 * Course: 19CSP01 - Free Open Source Software
 */
require_once "db.php";

$id = isset($_GET["id"]) ? (int)$_GET["id"] : 0;
$editing = $id > 0;
$error = "";

$task = [
    "task_title"   => "",
    "description"  => "",
    "due_date"     => "",
    "task_status"  => "Pending",
    "category_id"  => 1,
    "priority_id"  => 2
];
$subtasks = [];

if ($editing) {
    $stmt = $conn->prepare("SELECT * FROM tasks WHERE task_id = ?");
    $stmt->bind_param("i", $id);
    $stmt->execute();
    $task_res = $stmt->get_result()->fetch_assoc();
    if (!$task_res) {
        header("Location: index.php?msg=notfound");
        exit;
    }
    $task = $task_res;
```

```
    // Fetch subtasks
    $stmt_st = $conn->prepare("SELECT * FROM task_checklists WHERE task_id = ? ORDER BY checklist_id
ASC");
    $stmt_st->bind_param("i", $id);
    $stmt_st->execute();
    $subtasks_res = $stmt_st->get_result();
    while ($st = $subtasks_res->fetch_assoc()) {
        $subtasks[] = $st["item_title"];
    }
}

// Handle Form Submission
if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $title       = trim($_POST["task_title"] ?? "");
    $description = trim($_POST["description"] ?? "");
    $due_date    = !empty($_POST["due_date"]) ? $_POST["due_date"] : null;
    $status      = $_POST["task_status"] ?? "Pending";
    $category_id = (int)($_POST["category_id"] ?? 0);
    $priority_id = (int)($_POST["priority_id"] ?? 0);
    $new_subtask = trim($_POST["new_subtask"] ?? "");

    if ($title === "") {
        $error = "Task Title is mandatory and cannot be empty.";
    } elseif ($category_id <= 0) {
        $error = "Please select a valid Category.";
    } elseif ($priority_id <= 0) {
        $error = "Please select a valid Priority level.";
    } else {
        if ($editing) {
            // Update Existing Task
            $stmt = $conn->prepare("UPDATE tasks SET task_title = ?, description = ?, due_date = ?, task_status
= ?, category_id = ?, priority_id = ? WHERE task_id = ?");
            $stmt->bind_param("ssssiii", $title, $description, $due_date, $status, $category_id, $priority_id, $id);
            $stmt->execute();

            // Insert new subtask if provided
            if (!empty($new_subtask)) {
                $stmt_chk = $conn->prepare("INSERT INTO task_checklists (task_id, item_title) VALUES (?, ?)");
                $stmt_chk->bind_param("is", $id, $new_subtask);
                $stmt_chk->execute();
            }

            header("Location: index.php?msg=updated");
            exit;
        } else {
```

```
        // Create New Task
        $stmt = $conn->prepare("INSERT INTO tasks (task_title, description, due_date, task_status,
category_id, priority_id) VALUES (?, ?, ?, ?, ?, ?)");
        $stmt->bind_param("ssssii", $title, $description, $due_date, $status, $category_id, $priority_id);
        $stmt->execute();
        $new_id = $stmt->insert_id;

        // Insert optional subtask
        if (!empty($new_subtask)) {
            $stmt_chk = $conn->prepare("INSERT INTO task_checklists (task_id, item_title) VALUES (?, ?)");
            $stmt_chk->bind_param("is", $new_id, $new_subtask);
            $stmt_chk->execute();
        }

        header("Location: index.php?msg=created");
        exit;
    }
}

$task = [
    "task_title"   => $title,
    "description"  => $description,
    "due_date"     => $due_date,
    "task_status"  => $status,
    "category_id"  => $category_id,
    "priority_id"  => $priority_id
];
}

// Fetch categories and priorities for dropdowns
$categories = $conn->query("SELECT * FROM categories ORDER BY category_name ASC");
$priorities = $conn->query("SELECT * FROM priorities ORDER BY priority_level ASC");
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title><?= $editing ? "Edit Task" : "Create New Task" ?> - Campus Task Planner</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header class="app-header">
    <div class="header-container">
```

```
      <div class="brand">
        <h1> 📋 Campus Task Planner</h1>
        <p class="subtitle"><?= $editing ? "Edit Existing Task Record" : "Create New Relational Task" ?> |
24BCS174</p>
      </div>
      <div>
        <a href="index.php" class="btn btn-clear">⬅ Back to Dashboard</a>
      </div>
    </div>
</header>

<main class="container">
    <div class="form-container">
      <h2><?= $editing ? "✏️ Edit Task Details" : "➕ Add New Task Record" ?></h2>
      <p class="form-lead">Enter task details below. All entries are validated on the server and persisted into
relational tables.</p>

      <?php if (!empty($error)): ?>
        <div class="alert alert-danger">
          <strong>Validation Error:</strong> <?= htmlspecialchars($error) ?>
        </div>
      <?php endif; ?>

      <form method="POST" action="task_form.php<?= $editing ? '?id=' . $id : '' ?>" class="crud-form">
        <div class="form-group">
          <label for="task_title">Task Title <span class="required">*</span></label>
          <input type="text" id="task_title" name="task_title" value="<?=
htmlspecialchars($task['task_title']) ?>" placeholder="e.g. Complete FOSS Assignment 3 Report" required>
        </div>

        <div class="form-group">
          <label for="description">Task Description / Objective</label>
          <textarea id="description" name="description" rows="3" placeholder="Provide additional task
specifications or lab requirements..."><?= htmlspecialchars($task['description'] ?? '') ?></textarea>
        </div>

        <div class="form-row">
          <div class="form-group col-half">
            <label for="category_id">Category (Relational Table: categories) <span
class="required">*</span></label>
            <select id="category_id" name="category_id" required>
              <?php while ($cat = $categories->fetch_assoc()): ?>
                <option value="<?= $cat['category_id'] ?>" <?= $task['category_id'] ==
$cat['category_id'] ? 'selected' : '' ?>>
                  <?= htmlspecialchars($cat['category_name']) ?> (<?=
htmlspecialchars($cat['category_code']) ?>)
                </option>
              <?php endwhile; ?>
            </select>
          </div>
```

```
            <div class="form-group col-half">
                <label for="priority_id">Priority Level (Relational Table: priorities) <span
class="required">*</span></label>
                <select id="priority_id" name="priority_id" required>
                    <?php while ($p = $priorities->fetch_assoc()): ?>
                        <option value="<?= $p['priority_id'] ?>" <?= $task['priority_id'] == $p['priority_id'] ?
'selected' : '' ?>>
                            Level <?= $p['priority_level'] ?>: <?= htmlspecialchars($p['priority_label']) ?>
                        </option>
                    <?php endwhile; ?>
                </select>
            </div>
        </div>

        <div class="form-row">
            <div class="form-group col-half">
                <label for="due_date">Due Date</label>
                <input type="date" id="due_date" name="due_date" value="<?=
htmlspecialchars($task['due_date'] ?? '') ?>">
            </div>

            <div class="form-group col-half">
                <label for="task_status">Task Status</label>
                <select id="task_status" name="task_status">
                    <option value="Pending" <?= $task['task_status'] === 'Pending' ? 'selected' : ''
?>>Pending</option>
                    <option value="In Progress" <?= $task['task_status'] === 'In Progress' ? 'selected' : '' ?>>In
Progress</option>
                    <option value="Completed" <?= $task['task_status'] === 'Completed' ? 'selected' : ''
?>>Completed</option>
                </select>
            </div>
        </div>

        <div class="form-group">
            <label for="new_subtask">Checklist / Subtask (Relational Table: task_checklists)</label>
            <input type="text" id="new_subtask" name="new_subtask" placeholder="e.g. Verify MySQL
foreign key constraints">
            <?php if (!empty($subtasks)): ?>
                <div class="existing-subtasks">
                    <small><strong>Existing Checklists:</strong> <?= htmlspecialchars(implode(" • ",
$subtasks)) ?></small>
                </div>
            <?php endif; ?>
        </div>

        <div class="form-actions">
            <button type="submit" class="btn btn-save"><?= $editing ? "💾 Update Task Record" : "💾 Save
Task" ?></button>
            <a href="index.php" class="btn btn-clear">Cancel</a>
        </div>
    </form>
</div>
```

```
</main>

<footer class="app-footer">
    <p>Campus Task Planner | 19CSP01 Free Open Source Software | Ragul Ramachandran (24BCS174)</p>
</footer>

</body>
</html>
```

5. Task Deletion Handler Script (delete_task.php)

```
<?php
/**
 * delete_task.php - Delete Operation Handler
 * Student: Ragul Ramachandran (24BCS174)
 * Course: 19CSP01 - Free Open Source Software
 */
require_once "db.php";

$id = isset($_GET["id"]) ? (int)$_GET["id"] : 0;

if ($id > 0) {
    // Prepared statement to securely delete task (cascading removes checklists)
    $stmt = $conn->prepare("DELETE FROM tasks WHERE task_id = ?");
    $stmt->bind_param("i", $id);
    $stmt->execute();
}

header("Location: index.php?msg=deleted");
exit;
?>
```

Experimental Verification and Results

The full CRUD lifecycle of the web application was verified locally using the Apache Web Server and MySQL daemon:

A. Read Operation (Dashboard with Combined CRUD Actions)

Navigating to http://localhost/Assignment_3_TaskPlanner/index.php displays the styled task directory with real-time relational join data and combined CRUD action buttons:

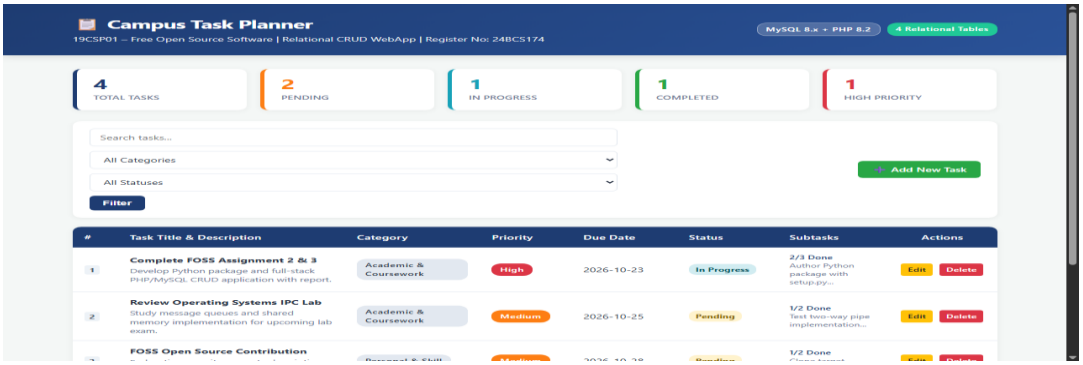


Figure 2: Home Page (Read Operation) Displaying Relational Data and Combined CRUD Action Buttons

B. Create Operation (Add Task Form and Success Notification)

Clicking the '+ Add New Task' button renders the registration form with dynamic category and priority selection dropdowns. Submitting valid data inserts the record and renders a green confirmation banner:

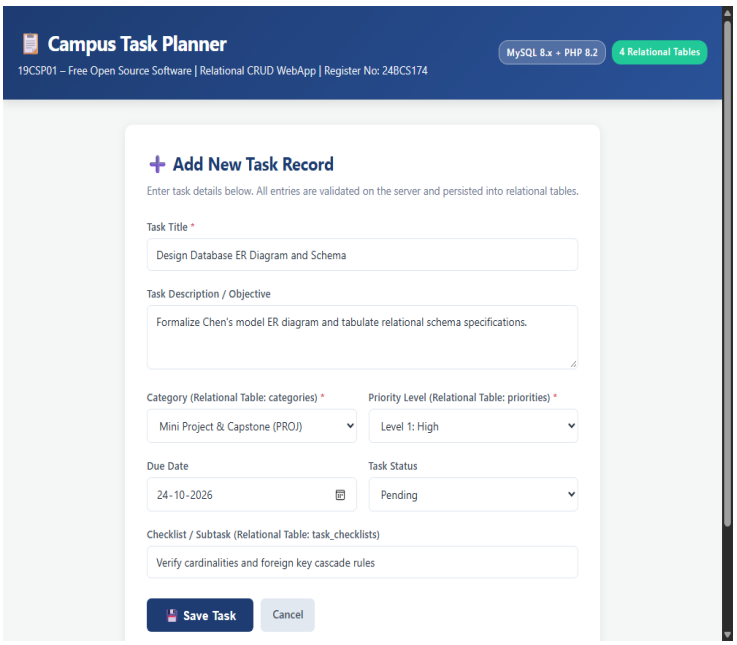


Figure 3a: Create Operation – Add Task Form with Dynamic Relational Dropdowns



Figure 3b: Create Operation – Successful Record Creation with Green Status Notification Banner

C. Update Operation (Edit Task Record)

Clicking 'Edit' on a task opens the pre-populated update form. Modifying attributes and clicking 'Update Task Record' saves the changes and displays an updated status alert:

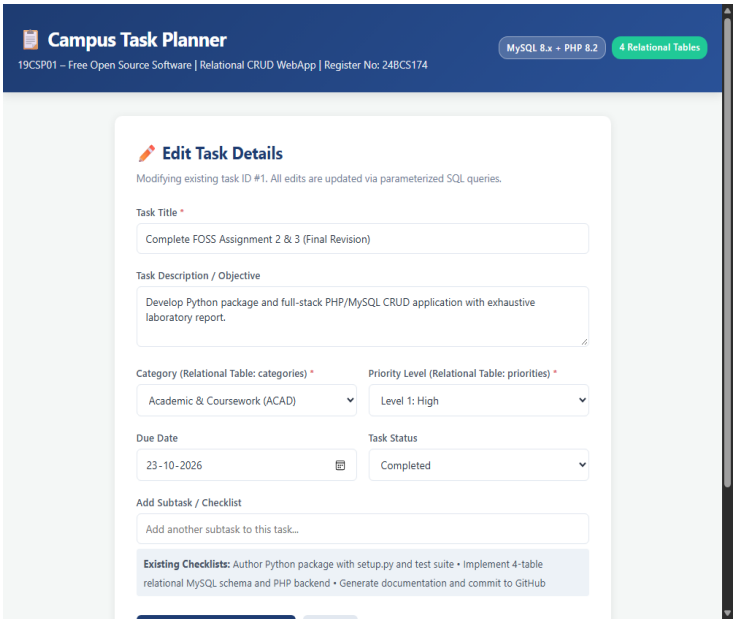


Figure 4a: Update Operation – Pre-populated Task Modification Form



Figure 4b: Update Operation – Successful Record Update with Status Notification Banner

D. Delete Operation (Task Record Removal)

Clicking the 'Delete' button triggers a JavaScript confirmation modal. Confirming execution removes the record from MySQL and immediately refreshes the dashboard with a red alert banner:



Figure 5: Delete Operation – Instant Red Alert Notification Banner Confirming Record Deletion

Summary of SQL CRUD Queries Executed

CREATE Query:

```
INSERT INTO tasks (task_title, description, due_date, task_status, category_id, priority_id)
VALUES ('Design Database ER Diagram and Schema', 'Formalize Chen\'s model ER diagram...', '2026-10-24',
'Pending', 2, 1);
```

READ Query:

```
SELECT t.task_id, t.task_title, t.description, t.due_date, t.task_status, c.category_name,
    p.priority_label, p.badge_color, GROUP_CONCAT(tc.item_title SEPARATOR ' • ') AS checklist_summary,
    COUNT(tc.checklist_id) AS total_subtasks, SUM(IF(tc.is_completed = 1, 1, 0)) AS completed_subtasks
FROM tasks t JOIN categories c ON t.category_id = c.category_id JOIN priorities p ON t.priority_id =
p.priority_id
LEFT JOIN task_checklists tc ON t.task_id = tc.task_id GROUP BY t.task_id ORDER BY t.due_date ASC,
p.priority_level ASC;
```

UPDATE Query:

```
UPDATE tasks SET task_title = 'Complete FOSS Assignment 2 & 3 (Final Revision)', task_status =
'Completed',
    category_id = 1, priority_id = 1 WHERE task_id = 1;
```

DELETE Query:

```
DELETE FROM tasks WHERE task_id = 3; -- Cascades deletion to task_checklists
```

Output / Result

A dynamic, full-stack Campus Task Planner web application was successfully designed and deployed using MySQL as the relational backend and PHP as the frontend. The system incorporates 4 relational tables (categories, priorities, tasks, task_checklists) linked via foreign keys. All four CRUD operations (Create, Read, Update, Delete) were comprehensively executed with responsive user feedback banners and strict server-side validation. All experimental workflows were verified on the local Apache and MySQL runtime.

Conclusion

This assignment provided comprehensive practical experience in developing full-stack database-driven web applications using foundational FOSS technologies. The synergy between PHP's server-side processing and MySQL's relational data integrity was demonstrated through the 4-table relational schema architecture, input sanitization, dynamic query construction, and responsive user experience design. The project underscores the effectiveness of open-source tooling in engineering scalable, enterprise-grade information systems.